

Vision Transformers

DATA 443/643 · Week 4, note 4.2

After Stanford CS231N 2025 L8: patches, a linear map, positions, no causal mask.

Lecture plan

- 1 Tile the image; do not use overlapping windows
- 2 Flatten, then a linear map to width d
- 3 Add positions; every patch may look
- 4 Start from a pretrained tower

First time: what / why / formula

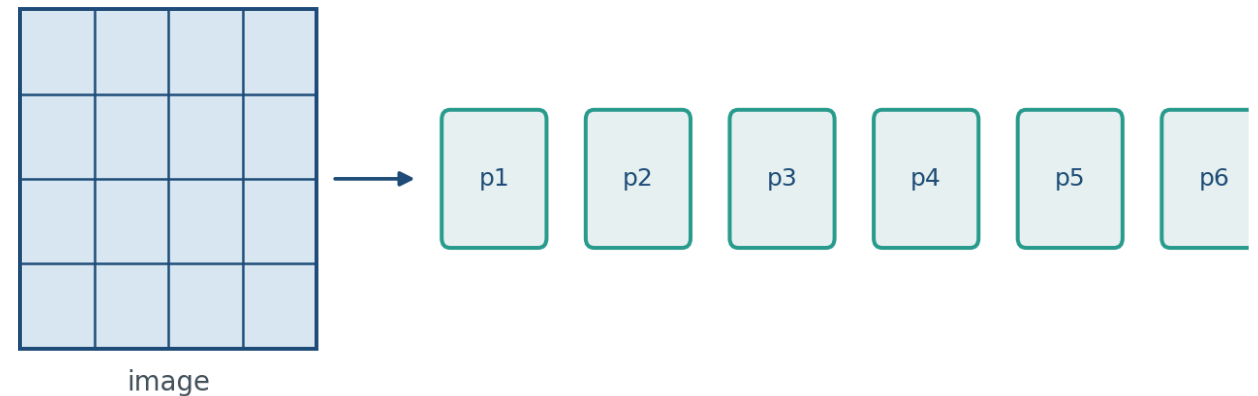
- What / why. Cut the image into non-overlapping patches, linearly map each to a d -vector, add positions, run a transformer without a causal mask. CNNs are the older default. ViT shows the same block as language can mix patches globally from layer 1.
- Architecture. Patch embed $E \in \mathbb{R}^{P \times d}$ plus optional CLS token, then encoder blocks. Classification: CLS or mean pool, then a head.
- Formula. For patch size $p \times p$ and image $H \times W$, sequence length $T = (H / p) \cdot (W / p)$. $X = \text{LN}(PW_e + E_{\text{pos}})$.
- Tradeoffs. + Global mixing, same code as NLP. – Quadratic in patches; needs data or regularization; patch size is a method knob.

How. Lab 4 flattens patches. No convolution is required for the cartoon. Pretrain on a large image set; small data prefers CNNs or a strong recipe.

Patches as tokens

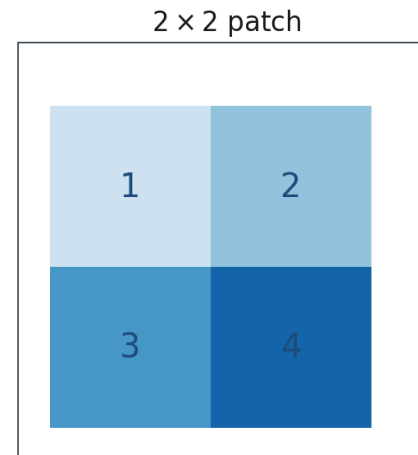
- $N = HW / P^2$ non-overlapping tiles.
- A 16 by 16 RGB patch is 768 numbers.
- That linear map is a conv with kernel P , stride P .

A ViT treats patches the way an LLM treats tokens

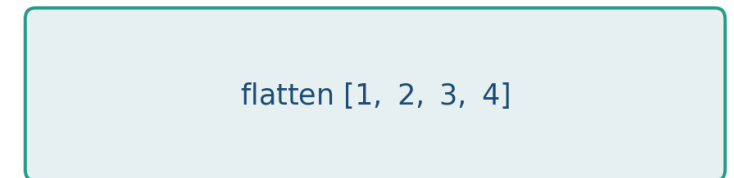


Flatten one tile

- A 2 by 2 of 1,2,3,4 becomes [1, 2, 3, 4].
- Then W maps to width d .
- Lab 4 does this on a 32 by 32 RGB crop.



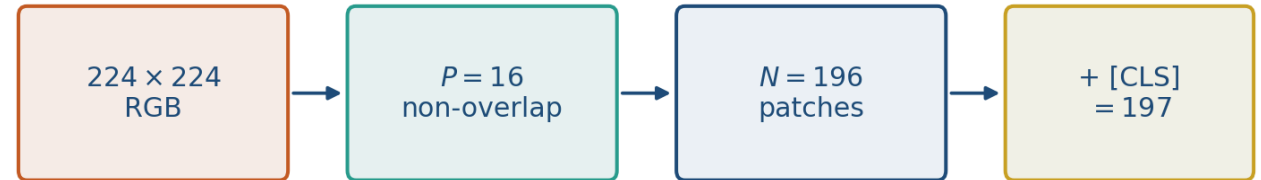
Then a linear map to width d



A 224 image is a short paragraph

- $P = 16$ gives $N = 196$, plus [CLS] is 197.
- Attention map: 197^2 scores per head.
- No GPT mask: there is no future patch.

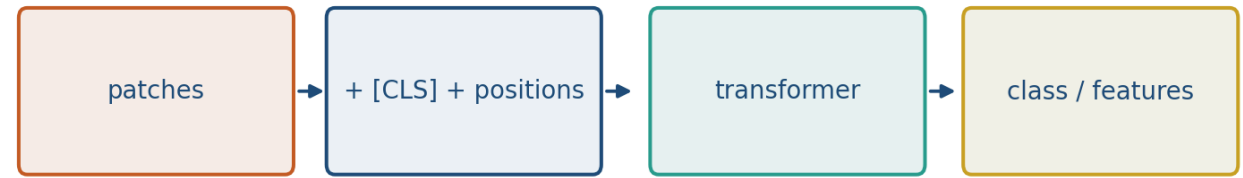
A 224 image is a short paragraph. No causal mask: every patch may look.



The same stack as language

- Positions are not optional. Top is not bottom.
- [CLS] or mean-pool, then a head.
- From scratch on 800 scans is the wrong default.

Vision transformer: same block as language, different tokens



What to take from this lecture

Count it

$N = HW / P^2$. Tiles, not CNN windows.

Reuse Week 3

Transformer block, bidirectional, add positions.

Freeze a tower

CLIP's image encoder is often this ViT, already trained.